

A Reinforcement-Learning Control Agent for the Greenhouse—Cucumber Digital Twin

Environment, Baseline, Agent, and the Value of Reactive Control

Greenhouses Project — Control-agent results

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Abstract

We formulate greenhouse climate control as a sequential decision problem on the validated digital twin (whose equations are documented separately) and ask whether a learned, weather- and crop-state-reactive policy can beat a well-tuned *constant* setpoint programme on **net profit**. We build the environment (a Gym-style daily-stepping wrapper of the twin with weather-year randomisation and a noisy forecast), a strong gradient-free baseline (cross-entropy optimisation of a constant programme), and a Soft Actor–Critic (SAC) agent. The central finding is that **the answer depends decisively on methodology**. Reactive control adds no value while the crop model makes the profit optimum nearly static; making the crop model realistic (a layered leaf-cohort canopy that responds to supplemental light) creates genuine structure. On that realistic twin, a naive RL setup (short training, selection on the tuning years, a corrupt weather year in the pool) makes SAC look *worse* than the constant programme — but a rigorous setup (clean data, disjoint train/validation/test weather-year splits, validation-based best-checkpointing, and enough training to pass a late learning “phase transition”) makes SAC **reproducibly beat** the constant programme: over 15 seeds it reaches a held-out TEST net profit of $+0.40 \pm 1.12 \text{ €/m}^2$ versus the constant programme’s -0.37 €/m^2 , beating it in **12 of 15 seeds**. We report both the naive failure and the rigorous success, because the contrast — rather than either number alone — is the methodological lesson. Parameter-inference details are out of scope.

1 The control problem as a Markov decision process

The agent chooses a daily climate programme to maximise the season’s net profit. Formally:

- **Horizon / episode.** One growing season of $N = 100$ days; one step = one day. A fresh weather year is sampled each episode (domain randomisation) from 31 fully-covered years of hourly Dutch weather (1988–2017, 2023; gap years 2018–2022 excluded automatically).
- **Observation** ($\in \mathbb{R}^{18}$): crop state (day-of-season, LAI, cumulative yield, node number, leaf mass, standing fruit load); a short climate summary (previous day mean air temperature, CO_2 , VPD); and a **noisy 3-day forecast** of the three control-relevant drivers (outside temperature, solar radiation, wind), each with lead-dependent error scaled by a “forecast-skill” knob (0 = perfect ... large = useless).
- **Action** ($\in [0, 1]^{10}$, decoded to physical setpoints): day and night temperature setpoints, CO_2 setpoint, lamp *duration* and lamp *start hour*, ventilation proportional band and offset, the screen-closing outside-temperature threshold, the VPD floor for humidity control, and the energy-screen radiation threshold. Planting **density** is an episode-level decision fixed at reset.
- **Transition.** One day of the closed-loop twin: the climate ODE is integrated for 24 h under the decoded setpoints (low-level PI/proportional controllers translate setpoints into actuators),

the day’s gross assimilate is computed and passed to the daily crop-growth update, and the resources spent (heating, CO₂, lamp electricity) are metered.

- **Reward.** That day’s **net profit** €/m²:

$$r_d = p \Delta Y_d^+ - q_h H_d - q_c C_d - (q_e^{\text{pk}} E_d^{\text{pk}} + q_e^{\text{off}} E_d^{\text{off}}) - \frac{F}{365} - [d=0] q_s \rho,$$

revenue from the day’s harvested fresh yield increment minus heating, CO₂, peak/off-peak lamp electricity, prorated fixed greenhouse cost, and (on day 0) the plant cost of the chosen density.

Economic coefficients (AGC-2024 / AGC-2018). cucumber price $p = 0.889$ €/kg; heating $q_h = 0.09$ €/kWh; CO₂ $q_c = 0.30$ €/kg; electricity $q_e = 0.30$ (peak 07:00–23:00) / 0.20 €/kWh (off-peak); fixed greenhouse $F = 15$ €/m²/yr; plant cost $q_s = 0.22$ €/stem. The resource accounting was validated against the real AGC-2018 Reference grower (CO₂ 0.088 vs 0.079 kg/m²/d; lamp electricity 1.18 vs 1.26 kWh/m²/d).

2 Baseline: cross-entropy optimisation of a constant programme

Because the twin is fast and deterministic, a strong non-learned baseline is a single *season-constant* programme optimised directly for mean profit. We use the cross-entropy method (CEM): a Gaussian over the 10 normalised setpoints plus density is iteratively refitted to its elite samples, each candidate scored as mean profit over four fixed evaluation years. CEM converges in ~ 12 iterations and provides both the “number to beat” and a warm-start for the agent.

3 The agent: Soft Actor–Critic

SAC is an off-policy maximum-entropy actor–critic well suited to continuous control. Our implementation (Julia/Flux) uses twin soft-updated Q -critics, a squashed-Gaussian policy with the tanh log-probability correction, automatic entropy-temperature tuning (target entropy – $\dim \mathcal{A}$), a replay buffer, and reward scaling. Two additions proved decisive:

1. **Lamp placement.** The action space includes the lamp *start hour* (not only duration), so the policy can steer supplemental light in time (e.g. into cheaper off-peak hours).
2. **Warm start.** The actor is behaviour-cloned to the CEM programme and the replay buffer is seeded with CEM rollouts, so the agent *improves on* the best constant programme rather than rediscovering it.

4 Results

4.1 Making the environment reactive: a crop-model prerequisite

On the original single-pool (“big-leaf”) crop the twin’s photosynthesis ignored lamp light, so supplemental lighting cost electricity and heat but produced no yield; the profit optimum was a nearly static “grow cold, lamps off” corner with almost no state dependence. Replacing the crop with the layered leaf-cohort canopy (which photosynthesises under lamp PAR and integrates light down the canopy) made lamps productive and created a genuine *interior* optimum. With the cohort crop, a no-lamp programme yields ~ 11 kg/m² at $\approx$ break-even, an 18 h-lamp programme yields ~ 33 kg/m² but loses ≈ -35 €/m² to electricity, and the CEM optimum sits between them at ~ 3 h of (mostly off-peak) lamps.

4.2 Reactive vs constant control (a single, favourable seed)

Table 1 compares the CEM constant programme with a *single* SAC training seed on the four CEM-tuning years, the 27 held-out years, all 31, and the single worst year (all at forecast-skill 0). On this seed SAC appears to win everywhere. Section 4.3 shows this does not survive replication.

Table 1: Mean net profit €/m² (higher is better) for one SAC training seed. The apparent cohort-twin win does *not* reproduce across seeds (Table 2); it is shown to document the single-seed trap. SAC cannot beat a tuned constant on the non-reactive big-leaf twin either.

Controller	tuned 4 yr	held-out 27 yr	all 31 yr	worst yr
big-leaf, CEM (constant)	−0.28	−1.85	−1.65	−42.8
big-leaf, SAC	−2.13	−3.64	−3.45	−44.6
cohort, CEM (constant)	−0.80	−2.19	−2.01	−46.2
cohort, SAC (duration only, cold)	−1.69	−3.28	−3.07	−47.8
cohort, SAC (placement + warm, <i>one lucky seed</i>)	−0.23	−0.74	−0.68	−35.9

On this one seed SAC appears to improve the held-out mean from -2.19 to -0.74 €/m² and the worst-year from -46.2 to -35.9 €/m² — a large apparent robustness gain. As the next section shows, this is a favourable draw, not a property of the method.

4.3 Rigorous evaluation flips the verdict

A first, *naive* replication — three seeds, 24k steps, selection on the tuning years, and a weather pool still containing a corrupt year — does not reproduce the single-seed win: it lands *worse* than the constant programme (Table 2). Taken alone this reads as “RL does not help,” but it conflates four separable defects, each of which we then remove.

Table 2: *Naive* multi-seed setup (flawed): 24k steps, selection on the tuning years, and a corrupt all-zero weather year still in the pool. It rightly refutes the single lucky seed, but its own flaws — not the algorithm — drive the poor numbers. Superseded by Table 3.

Split	CEM (constant)	SAC (3 seeds, mean $\pm$ std)
tuned 4 yr	−0.80	-0.63 ± 0.89
held-out 27 yr	−2.19	-2.56 ± 0.21
worst year	−46.2	-55.5 ± 10.3

The four defects and their fixes: (i) a **corrupt weather year** — 2023 in the record is all zeros (outside temperature and radiation = 0 for the whole season), costing *every* policy ~ -47 €/m² and dominating the “worst-year”; it passed a row-count coverage check and sat silently in the pool, and is now filtered out (usable years 31 $\rightarrow$ 30); (ii) **selection on the tuning years** rather than a held-out set; (iii) **no validation checkpointing**, so a late-training dip is reported as the result; and (iv) **too-short training**. The last is decisive: every seed’s validation curve crashes on early exploration, plateaus for tens of thousands of steps, then undergoes a **late learning “phase transition” around step ~ 30 – 33 k** and climbs sharply — 24k steps stop before it.

A **rigorous campaign** fixes all four: clean data; disjoint **TRAIN 18 / VAL 6 / TEST 6** weather-year splits (TEST never used for training or selection); **validation best-checkpointing** (keep the actor with best validation profit, from the warm-started policy onward, so a seed can never end below $\approx$ CEM); 60k steps with gentle exploration (lr 10^{-4} , low entropy target, no random warmup); and a stiff-solver fallback in the environment. Fifteen seeds, 3.9 h (`train_sac_long.jl`). The result (Table 3):

Table 3: Rigorous campaign, cohort twin, six held-out TEST years never used for training or selection. SAC reproducibly beats the constant programme and reaches positive net profit.

Controller	TEST net profit €/m ² (6 held-out years)
CEM (constant)	−0.37
SAC (15 seeds, best-val ckpt)	+0.40 ± 1.12 (median +0.29)
SAC beats CEM on TEST in 12/15 seeds; best seed +3.26 (beats CEM every TEST year); worst −1.02.	

SAC reaches a *positive* held-out profit of $+0.40 \pm 1.12 \text{ €/m}^2$ where the constant programme merely breaks even, beating it in 12/15 seeds; the $+0.77 \text{ €/m}^2$ gap is ≈ 2.7 standard errors and 12/15 is well above chance. In the few seeds whose training does not take off, best-checkpointing falls back to $\approx$ CEM. The reversal from the naive setup owes entirely to evaluation methodology and training length, not to any change of algorithm: on this problem a single RL seed, short training, and in-sample selection can each independently manufacture the wrong conclusion.

4.4 What the agent learned

Dissecting a trained policy (`diagnose_policy.jl`) refutes the intuitive hypothesis that the agent times lamps to daily weather: the correlation between chosen lamp hours and that day’s solar radiation is ≈ 0 ($-0.07, +0.05, +0.01$ over three years). Instead the policy performs **phenological light-budget allocation**, a pattern identical across years:

- *Establishment (days 0–25)*: lamps ≈ 0 — the small canopy cannot use the light.
- *Productive window (days ~ 30 –70)*: lamps 3–8 h, CO₂ dosed up, occasional warm pushes — the full canopy converts light into fruit with time left to ripen it.
- *Wind-down (days ~ 80 –100)*: lamps ≈ 0 — new fruit could not mature and be harvested before season end.

This policy’s mean lamp use (1.7–2.3 h) is *below* CEM’s constant 3.19 h: it spends *less* electricity but *places* it in the productive window — exactly the state-dependent behaviour a constant programme cannot express, and the source of the profit and generalisation gains in Table 3. (Confirming this same mechanism on the campaign’s best-validation actor saved to `sac_actor_best.jls` is a one-command follow-up.)

5 Discussion

The results answer “does RL help here?” with an important qualifier: *it depends on both the environment and the evaluation*. A learned reactive controller adds value only when the profit-optimal policy is genuinely state-dependent and the agent has the lever to act on it: on the lamp-blind big-leaf twin those conditions fail (a near-static optimum) and a constant programme is better; making the crop model realistic supplies the structure. Given that structure, the agent’s ability to actually *realise* a win proved just as dependent on *how it is trained and judged*. A naive protocol (short training, in-sample selection, a corrupt data year) makes RL look strictly worse; a rigorous one (clean data, a held-out test split, validation checkpointing, and enough training to pass a late learning phase transition) makes the same algorithm **reproducibly beat the strong constant baseline** — positive held-out profit, 12/15 seeds. The two together are the paper’s real lesson: on a low-dimensional, near-static economic optimum, a single RL seed or a short run can manufacture either a false win *or* a false failure, and only careful, held-out, multi-seed evaluation separates the two. The learned advantage is concrete and interpretable — phenological allocation of the supplemental-light budget across the crop’s developmental stages — which is why it also generalises to unseen weather years.

Limitations and next steps. Absolute profits are near break-even because the modelled Dutch autumn season and current tariffs are cost-dominated, so the swing from the constant programme to the learned policy, though reproducible, is modest in euros; richer tariff structure (dynamic pricing) and more actuator freedom would widen it. The win still carries ± 1.12 across seeds with $\sim 3/15$ seeds only tying CEM, so variance reduction is the main open problem: **demonstration-anchored replay** (a protected expert/CEM buffer oversampled in each minibatch), best-of- k or ensembled policies, and more seeds. Further directions: confirm the mechanism on the saved best actor; the forecast-skill ablation (is the forecast worth anything?); handing planting density and de-leafing to the agent; and a receding-horizon MPC over the twin as a second strong reactive baseline. For deployment today, the best-validation actor (or an ensemble of the top seeds) is the candidate controller, with the constant CEM programme as a safe fallback.